

Improving the prediction of high-risk transport refusals

A field checklist derived from linked prehospital records

Yachen Li

Supervised by Dr. David Buckeridge (McGill University) and Dr. Vianney Jouhet (Université de Bordeaux)

Transport refusal

After a 9-1-1 call, a patient with decision-making capacity may decline the transport the crew recommends.

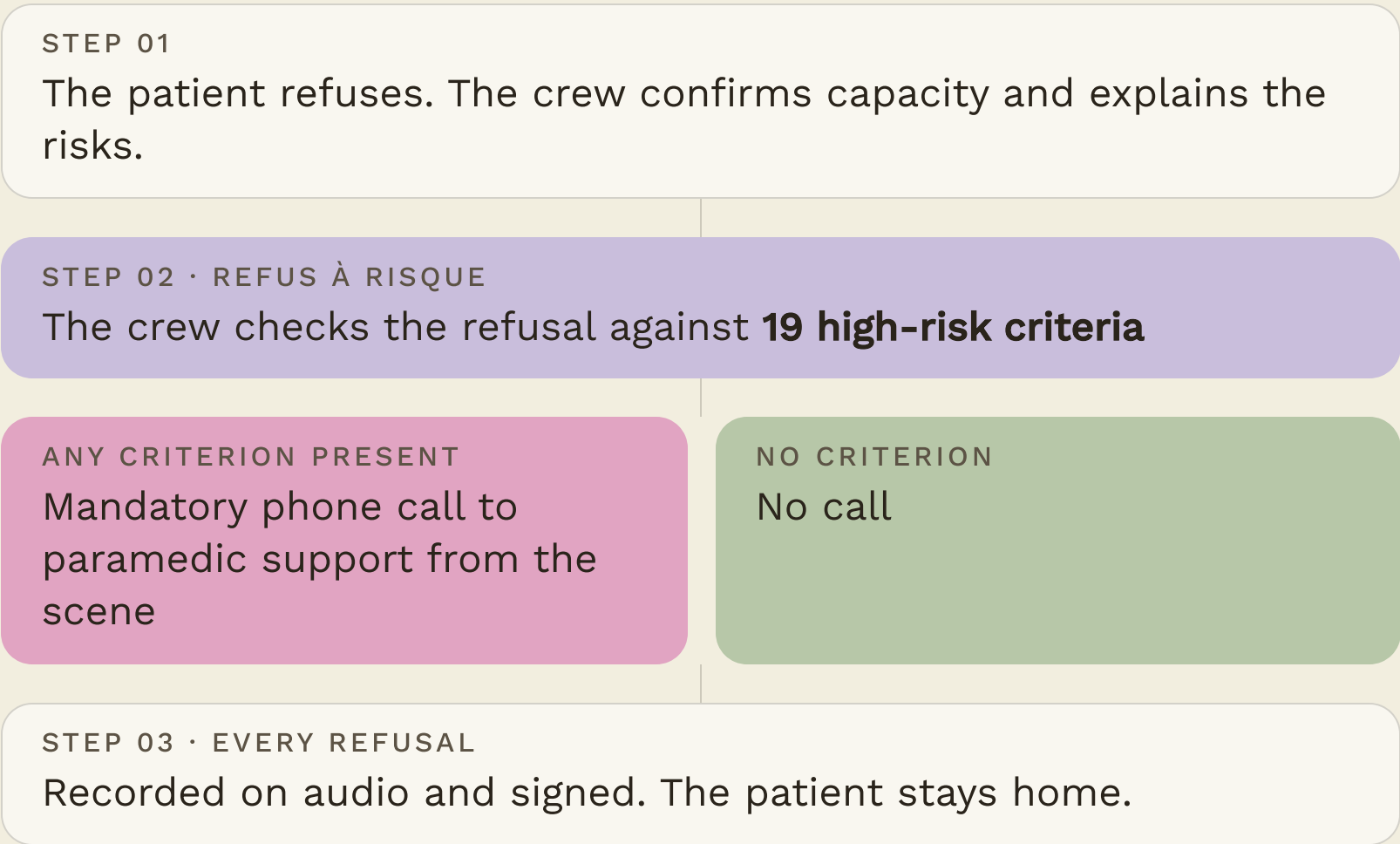
5–10%

of 9-1-1 calls in North American systems end with the patient refusing transport

6.3%

of non-conveyed patients were admitted to hospital within 72 hours, and 0.3% died

WHAT HAPPENS ON SCENE



A valid refusal must be respected. The task is to recognize the minority of refusals likely to end badly.

The *Refus à risque* criteria

Urgences-santé, Montreal and Laval

1	Under 14, no responsible adult on scene	2	Under 14 with any other risk criterion
3	Altered level of consciousness (V, P or U on AVPU)	4	New neurological deficit
5	Confusion, acute or chronic without a responsible adult	6	Breathing difficulty, reported or observed: SpO ₂ < 92%, cyanosis, retractions, wheezing
7	Non-traumatic pain under 48 h: headache, abdominal, back	8	Chest pain or palpitations
9	Pregnancy with abdominal pain, vaginal bleeding or syncope	10	Active hemorrhage
11	Hypoglycemia below 4 mmol/L	12	Intervention under the symptom-relief programme (PSS)
13	Pulse below 50 or above 120 per minute	14	Allergic reaction
15	Syncope or convulsions	16	Systolic BP < 100 or > 200 mm Hg, or diastolic > 100, interpreted according to age
17	Head trauma with signs or symptoms	18	Any trauma with a positive EQTPT score
19	Any trauma with a Glasgow Coma Scale of 14 or below		

Expert opinion. Not yet evaluated against what happened to the patient afterwards.

Objectives

01 · Evaluate

How well does the current protocol perform?
Evaluate the performance of the existing *Refus à risque* protocol.

02 · Build

A better checklist for paramedics
Develop an improved, paramedic-facing checklist for identifying patients at high risk of adverse outcomes after refusal.

03 · Benchmark

A ceiling for decision support
Develop a machine-learning model for risk stratification, to support clinical decision-making.

Design, data and cohort

DESIGN · SETTING · PERIOD

Retrospective cohort
Urgences-santé, Montreal and Laval
January 2015 to December 2022

THREE LINKED DATA SOURCES

- 1 Computer-aided dispatch
the 9-1-1 call and its closure
- 2 Paramedic runsheet
what the crew recorded on scene
- 3 Provincial health records
hospitalization and death afterwards

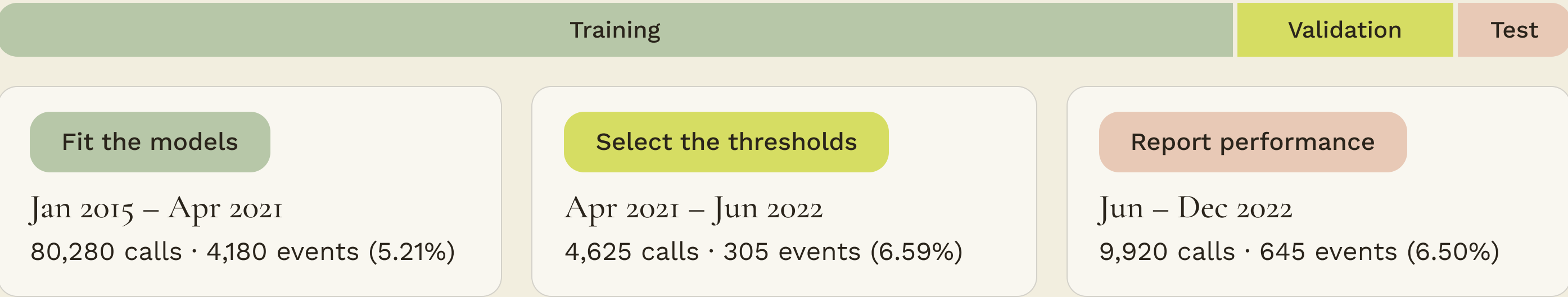
COHORT



Linkage to health records succeeded for 56% of refusals; unlinked refusals were excluded.

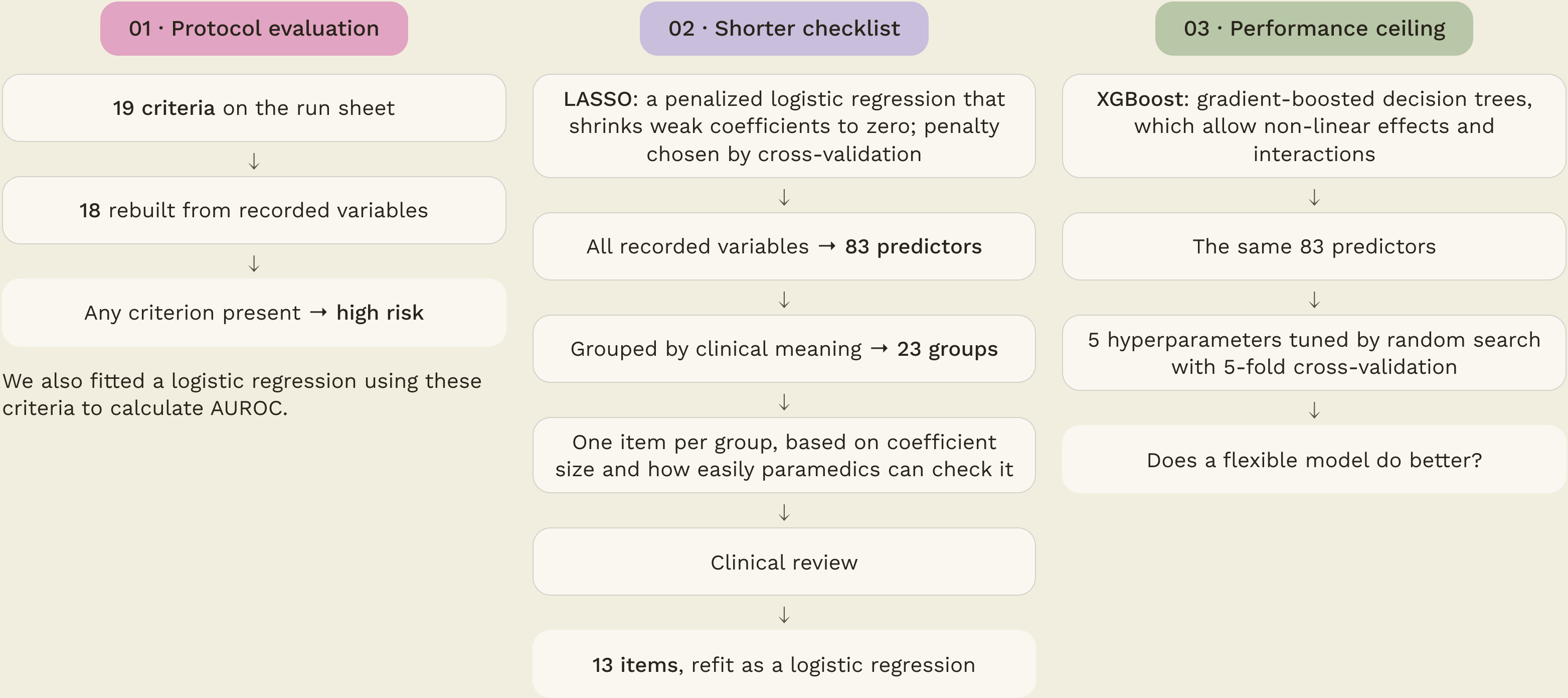
Chronological data split

Models were built on earlier calls and tested on later ones. The pandemic period was divided at its midpoint.



Analytic approach

Three analyses, all scored on the same test-set calls.



Model evaluation

Every measure is reported on the test set: June to December 2022, 9,920 refusals.

Operating point

Thresholds matched to the protocol

Each model's threshold was set on the validation set to reach the protocol's own sensitivity, then 85% and 90%. Sensitivity, specificity, PPV and NPV were then measured on the test set.

Discrimination

Does the rule rank risk correctly?

AUROC, with the area under the precision–recall curve as a check because only 6.5% of refusals had the outcome.

Calibration

Are the predicted risks the right size?

Calibration curve, slope and intercept, and Brier score.

Clinical utility

Are decisions better with the rule?

Decision curve analysis: net benefit compared with the current protocol, treating everyone and treating no one.

Performance of the current protocol

Refus à risque criteria applied to the 9,920 test-set refusals

Refusals flagged	75.8%
Sensitivity	77.7%
Specificity	24.4%
Positive predictive value	6.7%
Negative predictive value	94.0%

Most alarms are false, and about one in five patients who were later hospitalized or died is missed.

The 13-item checklist

Odds ratios from the refitted logistic regression.

Vital signs measured on scene		Presenting complaint at dispatch	
Tachycardia · heart rate > 100/min	1.81	Unconscious or fainting †	0.79
Tachypnea · respiratory rate > 22/min	1.51	Breathing difficulty	1.35
Hypoxia · SpO ₂ < 94%	1.88	Cardiac problem or implanted defibrillator †	0.63
Hypertension · systolic BP > 180 mm Hg †	0.71	Signs of stroke or transient ischemic attack	1.44
Patient background		Psychiatric presentation or suicide attempt	
Older age · per 10 years	1.39	General appearance	1.63
Medical history or current medication	2.24	Abnormal general appearance or skin colour	1.33
Alcohol or drug intoxication †	0.53		

† Kept on clinical grounds despite an odds ratio below 1: signs that are dangerous to miss.

Comparison of the four rules

Thresholds set on the validation set to match the protocol's sensitivity.

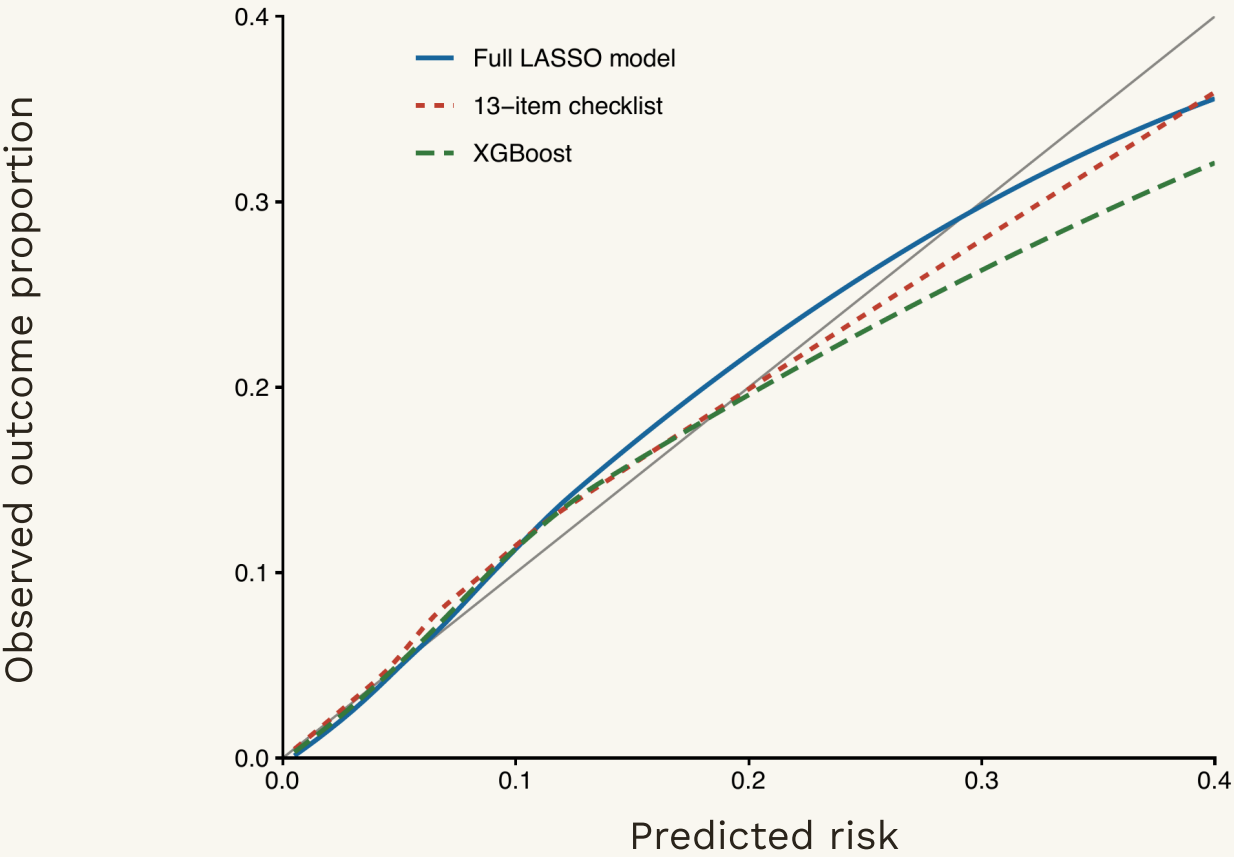
	Current protocol	13-item checklist	Full LASSO model	XGBoost
Refusals flagged	75.8%	46.3%	43.3%	40.0%
Sensitivity	77.7%	83.7%	82.8%	80.5%
Specificity	24.4%	56.4%	59.4%	62.8%
Positive predictive value	6.7%	11.8%	12.3%	13.0%
AUROC	0.619 [*]	0.758	0.772	0.783

^{*} For AUROC, we fitted a separate logistic regression using the protocol criteria. The other values show how the protocol performs when any one criterion triggers a flag.

For every 1,000 refusals, the checklist triggers about **300 fewer calls** and misses about **4 fewer patients** who were later hospitalized or died.

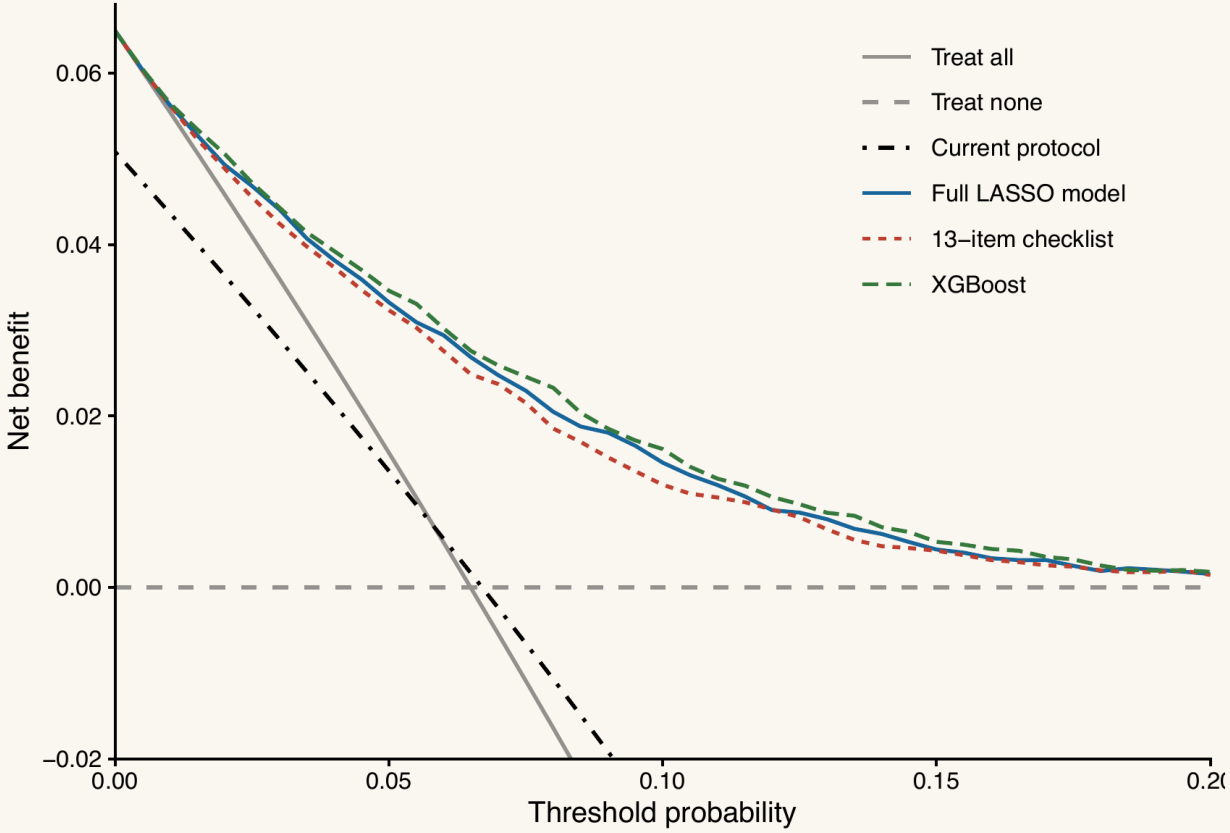
Calibration and clinical utility

Calibration



The checklist's predicted risks were close to the observed risks overall (slope 0.97, intercept 0.01).

Decision curve



All three models had a higher estimated net benefit than the protocol at thresholds from 0 to 0.20.

Clinical implications

In Montreal and Laval, the alarm is a phone call.

Today

Low specificity, sensitivity not high enough

Specificity 24%, sensitivity 78%: most alarms are false, and one in five at-risk patients is still missed.

Why

Risk after refusal is not acuity

Severely ill patients are usually transported. Among those who do refuse, what sets the high-risk ones apart is not their vital signs but their age, their medical history and why they called.

What changes

The threshold can be chosen

With a checklist, the service decides how many items trigger the call: at the protocol's sensitivity, 300 fewer calls and 4 more at-risk patients per 1,000 refusals.

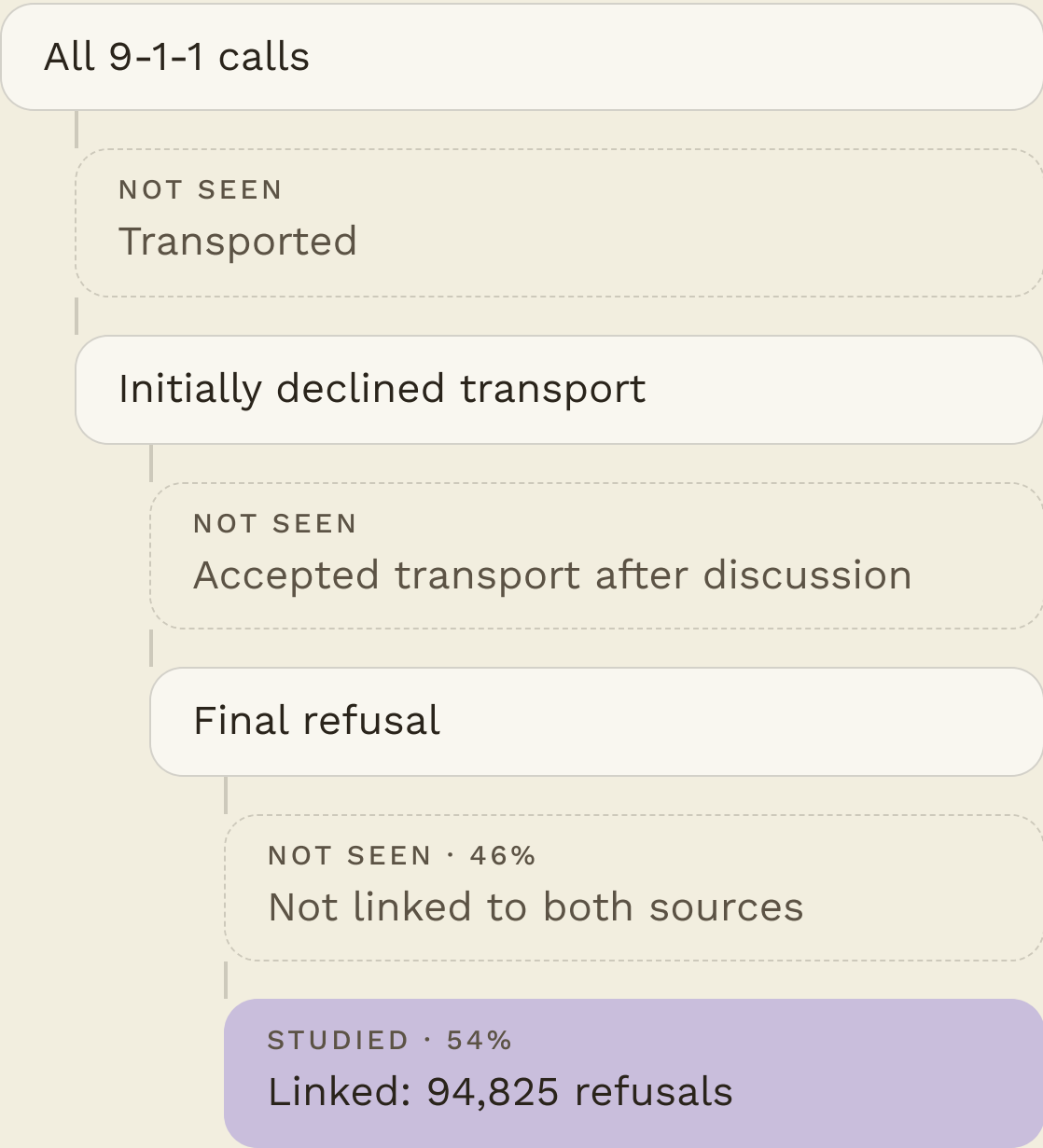
In practice

No extra work for the crew

Every item is already available to the crew: a paper checklist at the scene, the XGBoost model where the infrastructure exists.

Limitations inherent to the question

What the data can and cannot see

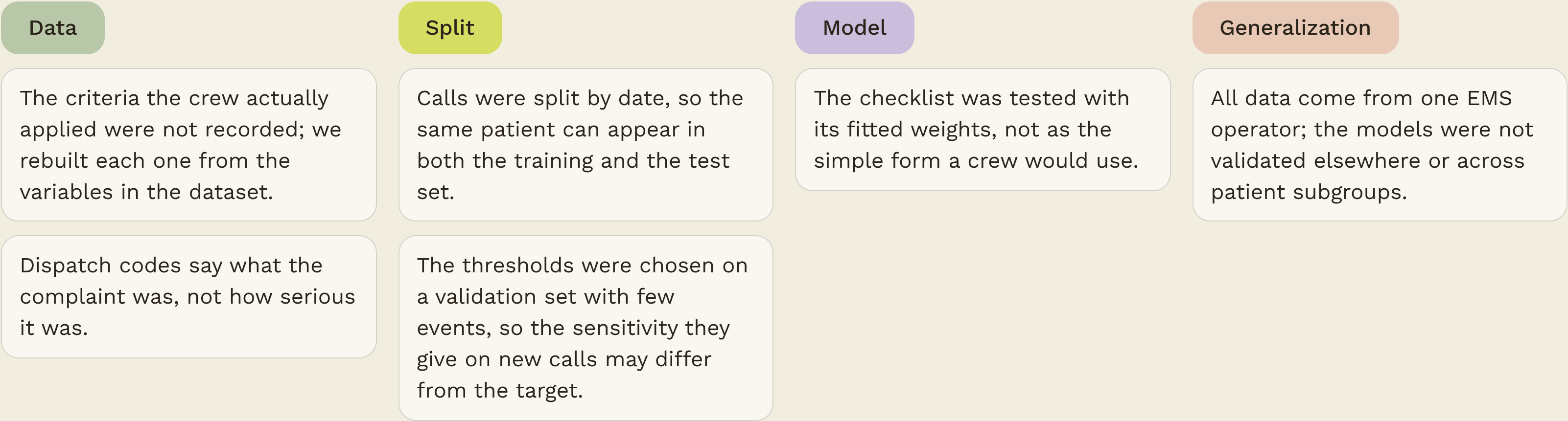


Refusers are a selected group. Danger signs that usually lead to transport are rare among them, and some carry negative weights. Neither means the sign is safe.

Patients who were dissuaded appear as transports. The protocol was judged only on the refusals it did not prevent, so its true performance may be better than reported.

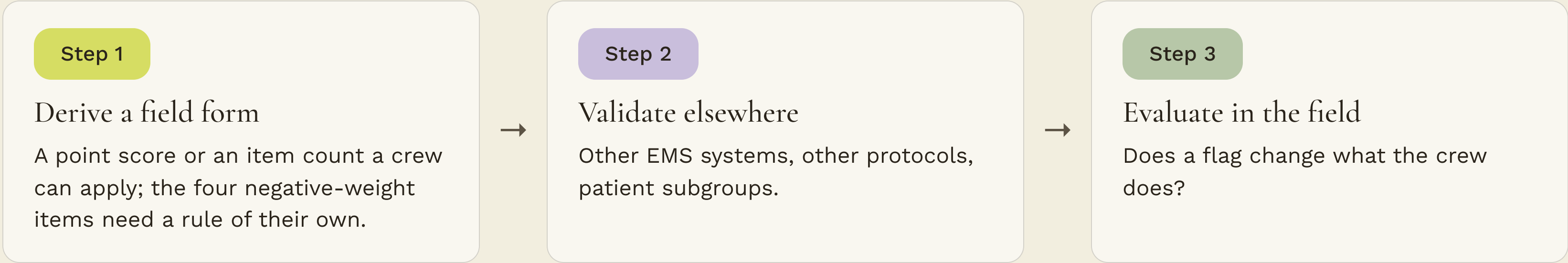
Half of the refusals could not be linked. The patients we could not see may be the ones at higher risk.

Limitations of the data and methods

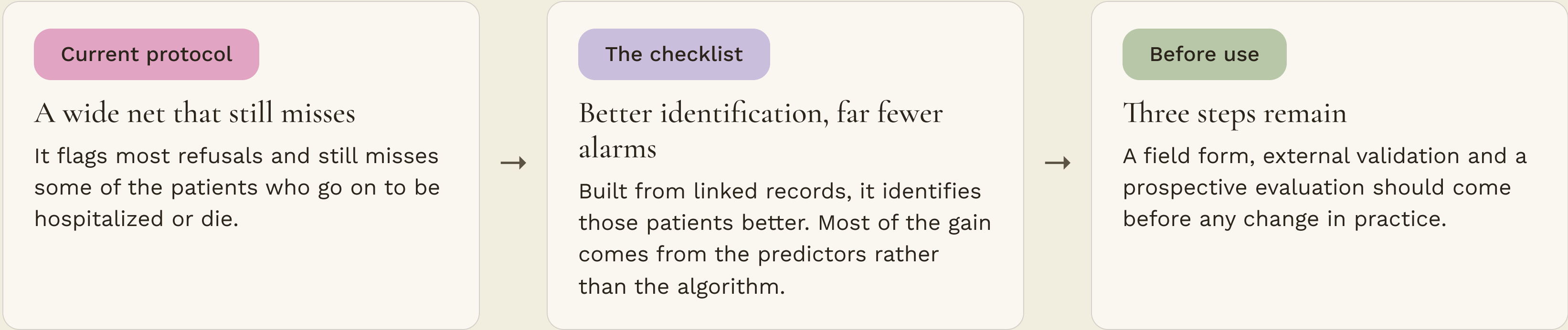


Perspectives

Three steps before any change in practice.



Conclusions



Thank you

Supervision

Dr. David Buckeridge
Dr. Vianney Jouhet
Dr. Luc de Montigny

Data

Dr. Aman Verma
Surveillance Lab · Urgences-santé
CADRISQ

Programme

ISPED, Université de Bordeaux
McGill University